MM141 Lab

Atomic Force Microscopy

Presented by: Muhammad Hamza Mahmood

Introduction to Materials and Nanotechnology Lab

Other Microscopy Techniques

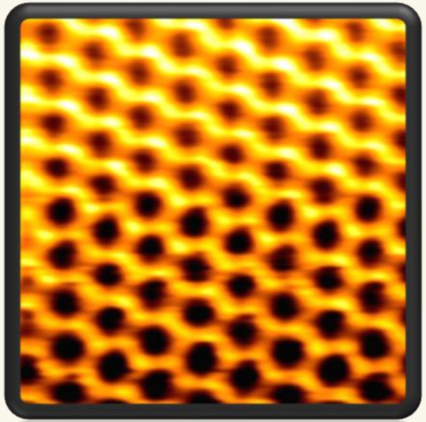
- Optical Microscopy
- Scanning Electron Microscope
- Transmission Electron Microscope

KEY TAKEAWAYS FROM AFM

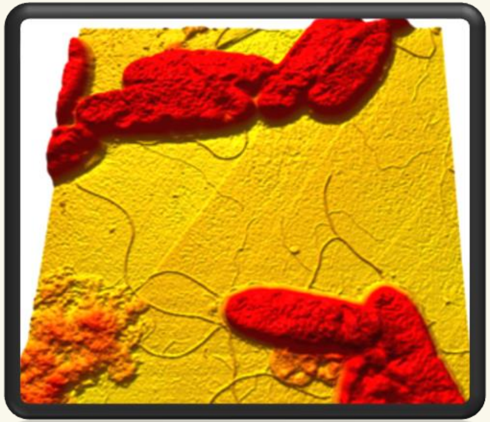
- **Topographic images** (3D surface maps)
- **Surface roughness** (e.g., Ra, RMS)
- **Mechanical properties** (e.g., stiffness, elasticity, hardness)
- **Phase imaging** (material phase distribution)
- **Electrical properties** (e.g., surface conductivity, conductive AFM)
- **Magnetic properties** (e.g., magnetic force microscopy)

Results of Atomic Force Microscopy

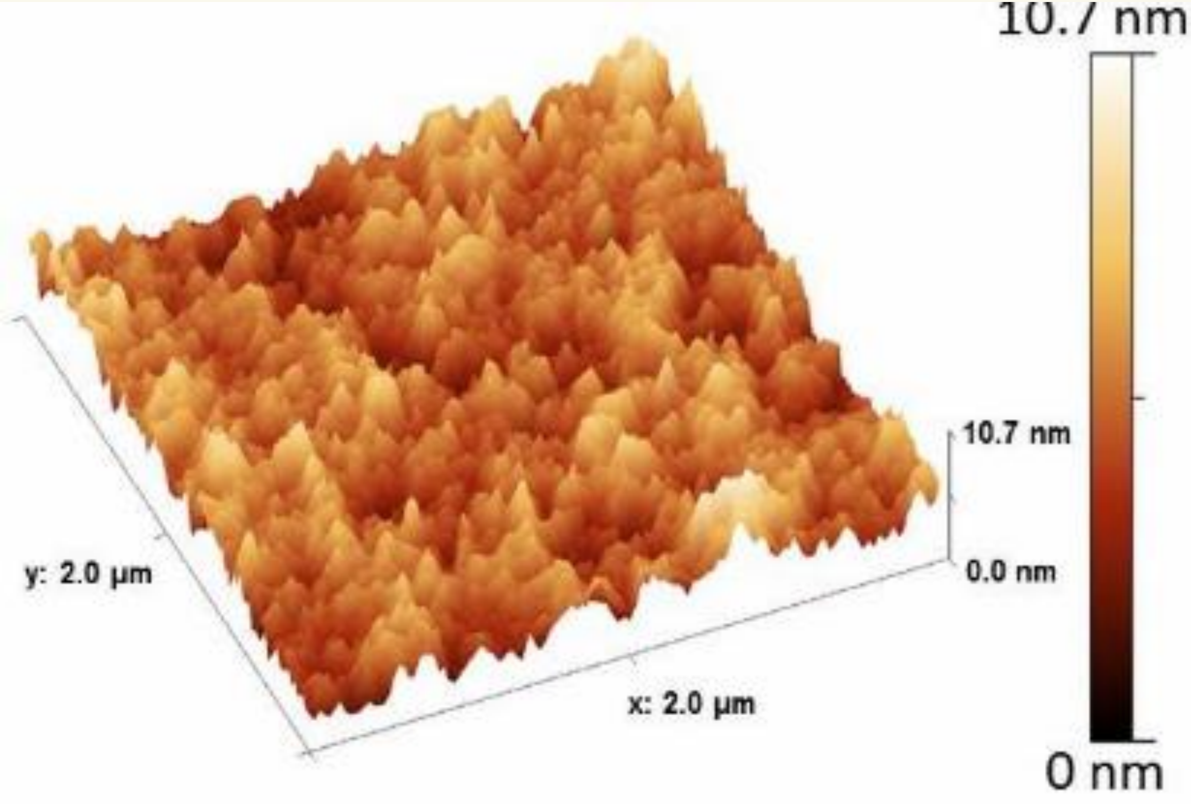
Graphite



Bacteria

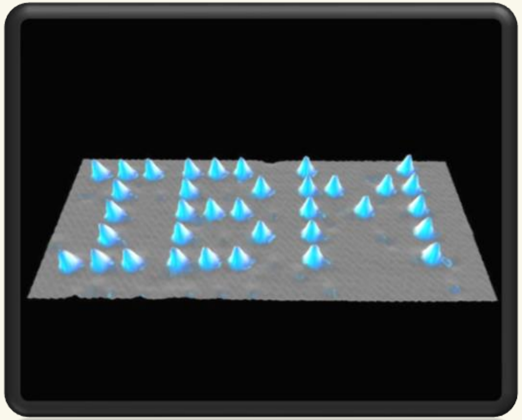


Nano-contact Printing

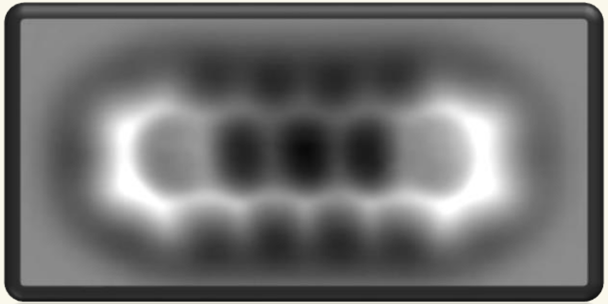

10.7 nm
10.7 nm
0.0 nm
y: 2.0 µm
x: 2.0 µm
0 nm

Results of Atomic Force Microscopy

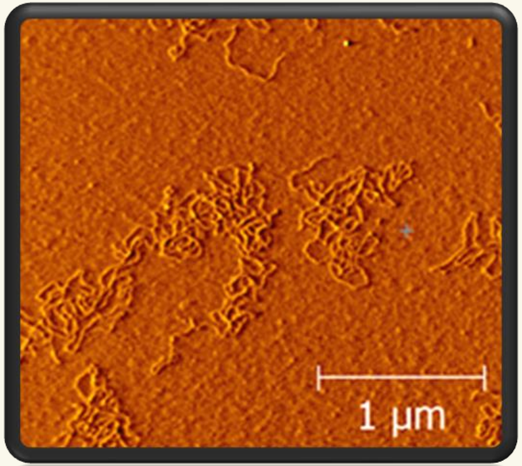
IBM writing



Pentacene



DNA

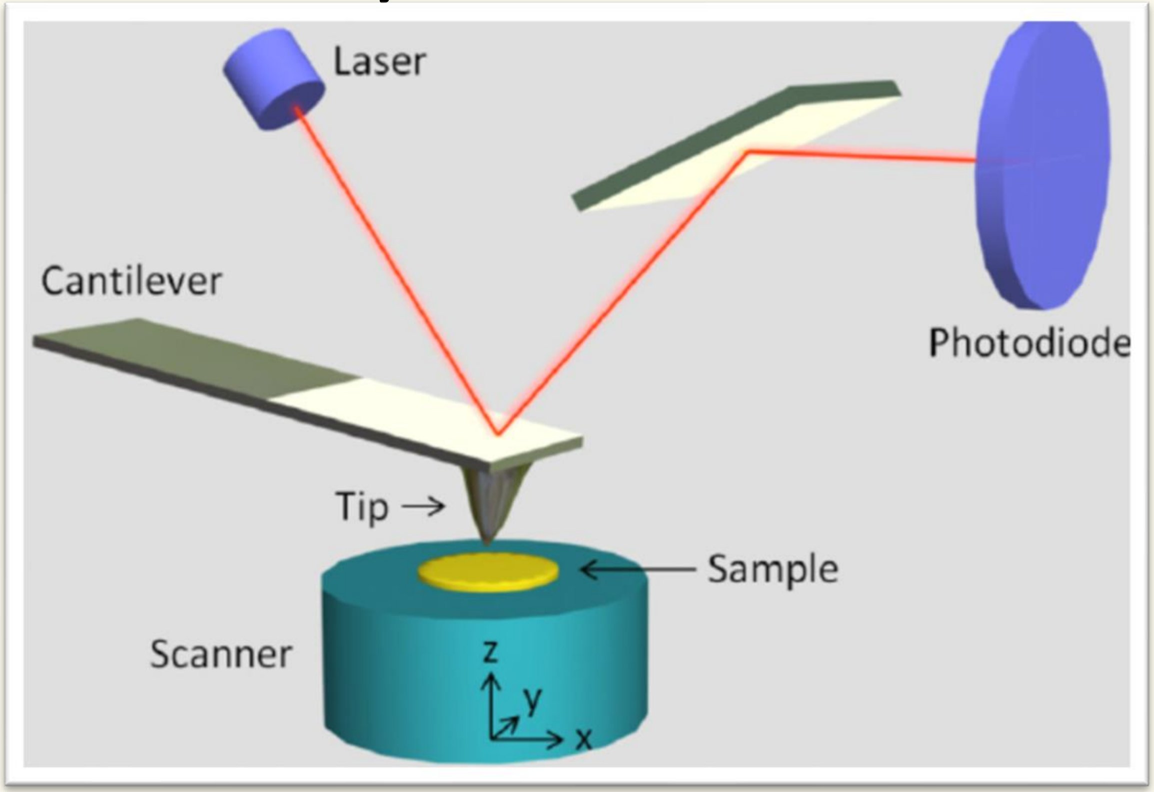


AFM

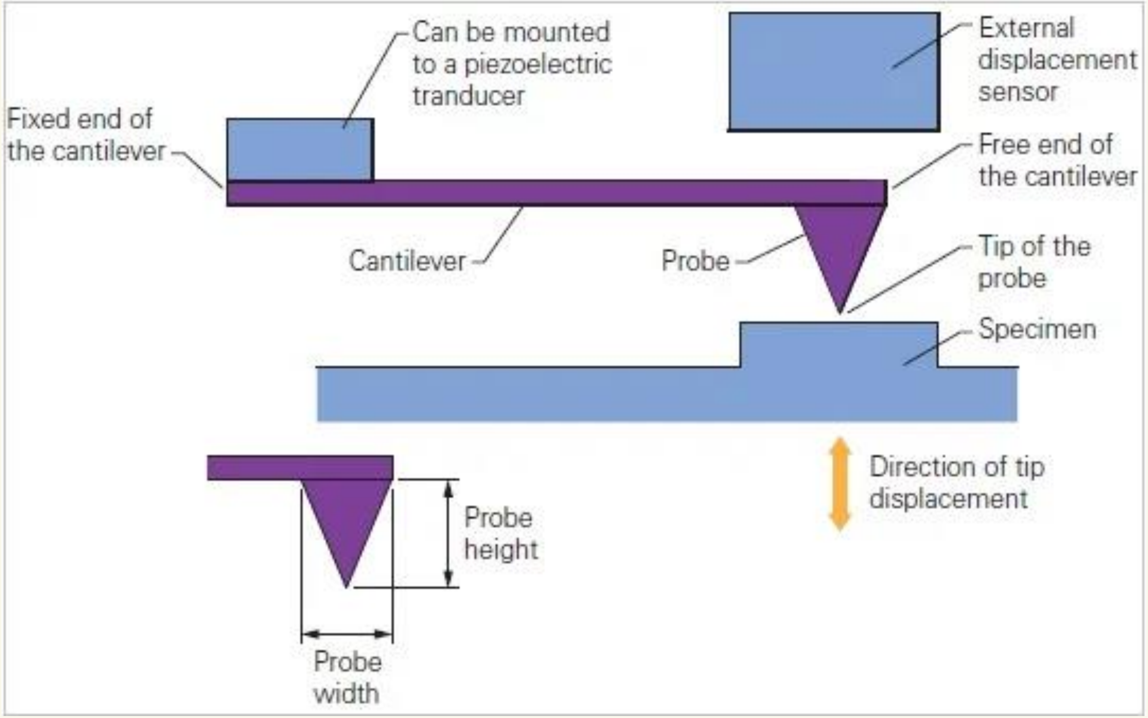
Atomic force microscopy (AFM) is one kind of observation method that takes advantage of the atomic force between the probe and the sample surface.

- Characterization Tool

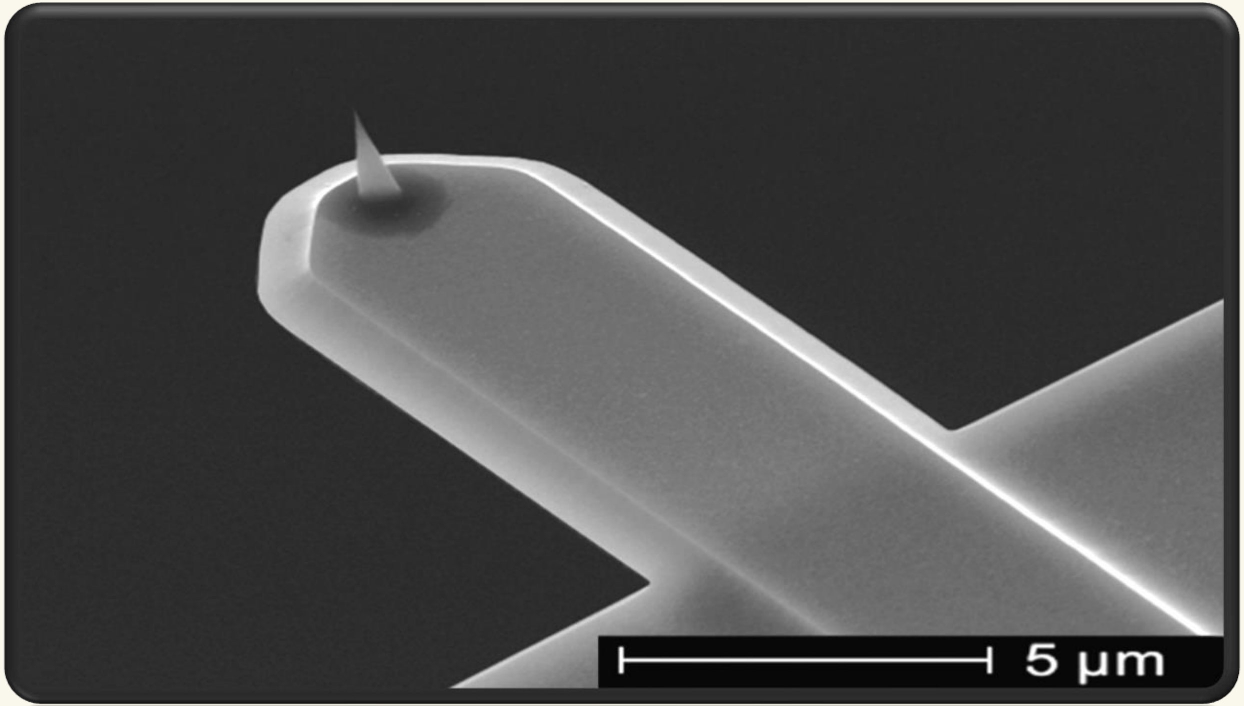
Components of AFM



Side view of Cantilever



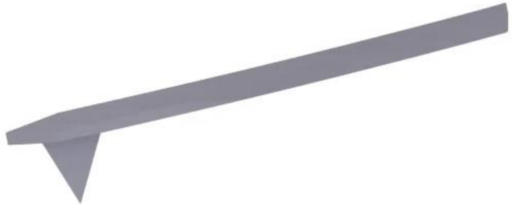
Tip of AFM



Understanding the working of AFM

- AFM is a type of Scanning Probe Microscopy that uses a very sharp probe to scan the surface of an object to be imaged.
- As the probe raster scans the surface, any deflection from its natural position due to the presence of surface features is detected.
- This deflection data is plotted in a computer to produce an image of the surface, allowing for high-resolution imaging of the object at the nanoscale.

Park SYSTEMS



Contact Mode

HOW AFM WORKS

Types of forces involved in AFM

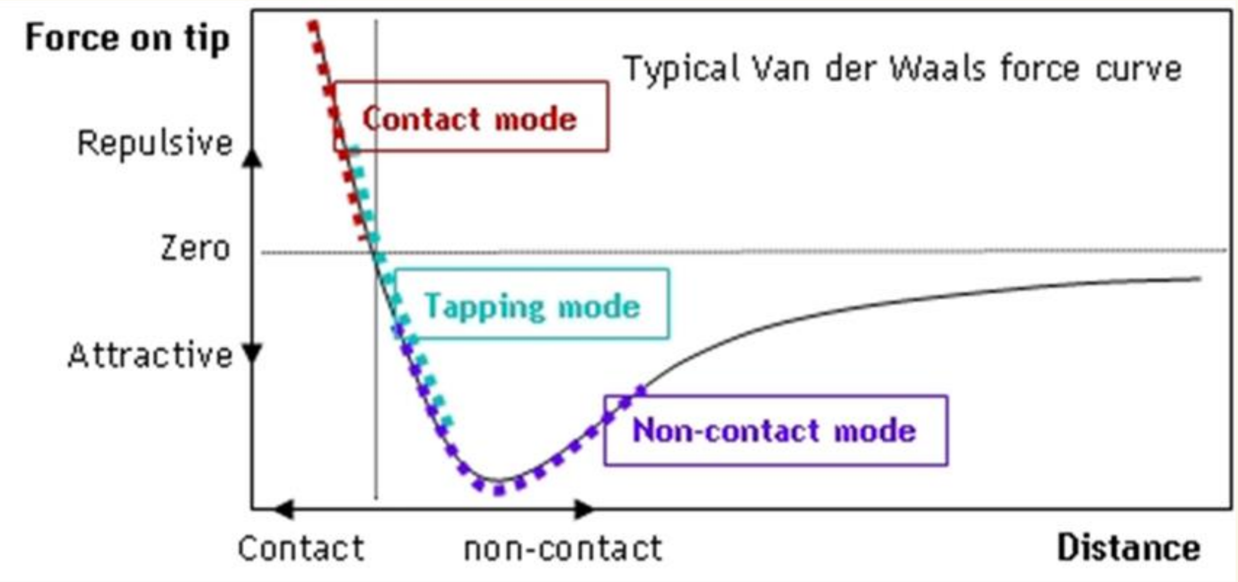
1. Near Field Forces

2. Van der Waals Forces

3. Repulsive Forces

Interaction between Tip and Surface

Force



Distance

AFM can operate in various modes including

Operation Modes

Contact Mode

Probe is continuously in physical contact with the sample surface during scanning

Non-Contact Mode

Probe oscillates near the sample surface without making physical contact

Tapping Mode

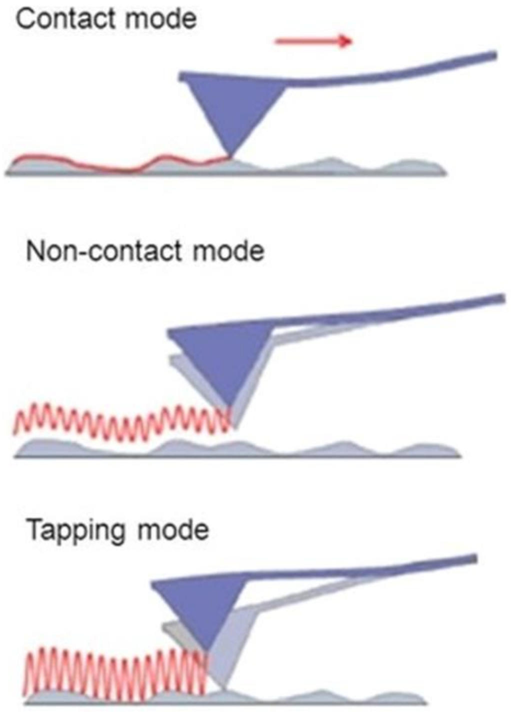
The probe oscillates at the resonance frequency, intermittently tapping the sample surface to minimize damage and enable high-resolution imaging of delicate samples.

Modes of operation. There are 3 modes of AFM operation

1. Contact mode

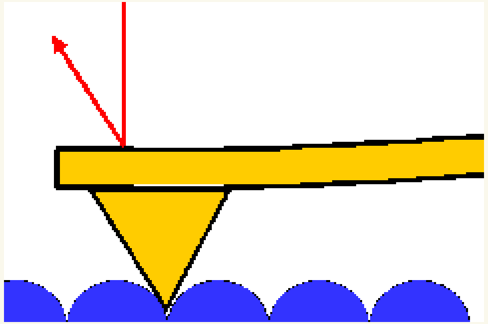
2. Non-contact mode

3. Tapping mode



Contact Mode

- AFM tip is brought close to the sample surface.
- Repulsive near-field forces act on the tip.
- High-resolution 3D images of the object are produced.



Pros

- High-Resolution Images
- Straightforward Circuit is involved.

Cons

- Tip gets damaged over time or blunt
- Soft and Delicate samples like Polymers, and biological tissue may get damaged

Non-Contact

- AFM tip does not touch the sample surface.
- Cantilever oscillates at its resonant frequency.
- Van der Waals forces decrease the resonant frequency and amplitude of the tip.
- Tip is moved away to maintain amplitude and frequency.

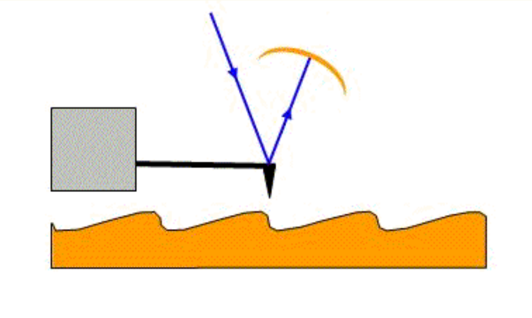
Pros

- Doesn’t damage the sample
- Used for biological samples and organic Thin films

Cons

- The non-contact mode can be a time-consuming process

Tapping Mode



- Tip just taps over the sample surface
- Tip is vibrated at the resonant frequency.
- Frequency and Amplitude is maintained constant
- When the tip nears the raised feature, the amplitude of the vibration

Pros

- It does not suffer from poor force detection on the sample due to the presence of moisture on the sample surface.
- Good Surface Resolution
- Minimal Surface Damage
- Gives you the idea about adhesion, viscosity, and friction

AFM

Advantages

- 3-D Image
- Non-conductive samples can be imaged
- Imaging can be done in liquids
- Higher Resolution than SEM
- No need for a vacuum
- Living molecules can be studied

Disadvantages

- Tips get blunt and need to be replaced
- Small Scan area compared to SEM
- A large number of Artifacts
- Very Slow Process
- Very Prone to vibrations in the environment.